

DRAFT — Design Basis Accident — Military Action: An Introduction to the Framework

Design Basis Accident — Military Action

An Introduction to the Framework and Document Architecture

Version 1.9 — DRAFT

April 2026

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Preface

This framework did not begin at a desk. It began in Ukraine.

Over the course of nearly a decade, I made seven trips to Ukraine to work alongside their security and critical infrastructure professionals on a complex and evolving set of challenges — grid resilience, cyber defense, incident response, and the coordination structures that hold together under pressure and those that do not. Those engagements gave me something that no amount of analysis from a distance can substitute for: a ground-level understanding of how a country with capable, dedicated people and genuinely critical infrastructure manages its vulnerabilities in real time, under real constraints, with real adversaries already probing the perimeter.

By late August 2021 — roughly five months before the invasion — I realized it was important enough to gather a small analytical team to start working on the problem seriously. The team, a HYDRA group, included a power systems engineer, Jeff R., an ICS/OT intel specialist, Rob L., and a well-known cyberattack planner and IT/OT expert, Mark F., among others in the State Department. Our focus was specifically on the threat to Ukrainian critical infrastructure: what an escalating Russian campaign might target, how the systems would respond, and where the critical vulnerabilities lay.

By early 2022, it was clear to me that the situation was moving toward something far more serious than a continued low-grade campaign. On February 6th of that year — eighteen days before Russian forces crossed the border — a letter I had written to Ukrainian authorities laying out what I believed was coming was delivered to UA officials at the US Embassy. Among the concerns that had been troubling me most: what would happen to Ukraine's nuclear plants if military forces moved on them? Ukraine operates fifteen commercial reactors across four sites. Zaporizhzhia alone, Europe's largest nuclear power plant, holds six. The existing frameworks for nuclear safety and physical security had not been written with that question in mind.

Zaporizhzhia answered the question in March 2022, when Russian forces seized the plant under active combat conditions. It has continued answering it every day since. As of this writing, the plant has been under military occupation for more than three years — a situation without precedent in the history of commercial nuclear power. What became undeniable, watching that situation unfold, was not merely that the Design Basis Threat established for commercial nuclear facilities had been exceeded. The deeper problem was that no formal analytical methodology existed for the scenario unfolding in real time. The design basis framework told engineers, operators, and regulators precisely what to do when a pump failed, a pipe broke, or a seismic event exceeded a threshold. It told them nothing about what to do when an armed force occupied the control room, targeted the diesel generators, and denied access to the site for months on end.

That gap is what this series is designed to close. The methodology developed here is built as a rigorous extension of the existing frameworks — the IAEA safety standards, the NRC licensing basis structure, the defense-in-depth philosophy that has guided commercial nuclear safety for

decades — not as a critique of them. Those frameworks were developed by capable people making considered judgments about the credible threats that commercial nuclear facilities faced. Those judgments were reasonable when they were made. The experience of Zaporizhzhia established that they are no longer sufficient as a complete answer. This document and the series it introduces constitute the analytical response to that recognition.

1. Purpose of This Document

The gap this document addresses — the absence of any formal methodology for assessing and mitigating the consequences of deliberate military action against a nuclear facility — is not a theoretical construct. It is an operational reality that has been unfolding at Zaporizhzhia for more than three years, and it demands an analytical response grounded in the same disciplined engineering methodology that has defined nuclear safety since its inception.

The technical documents in the DBA-MA series are written for nuclear engineers, safety analysts, regulators, and defense planners who will work directly with the methodology. This document is written for everyone else who needs to understand what the series is for, why it is structured the way it is, and what it is trying to accomplish before they can engage productively with the technical material.

It proceeds in four steps. First, it describes how nuclear facilities actually stay safe — the relationship between engineered design, defined threat envelopes, and disciplined human operation that constitutes the existing safety and security framework. Second, it identifies precisely where that framework breaks down when military action is the initiating condition. Third, it describes how the DBA-MA series is structured to close that gap. Fourth, it explains how the Transition Architecture addendum manages the period between the current posture of existing facilities and full DBA-MA compliance, because the gap cannot be closed overnight, and the transition itself must be managed analytically.

Readers who complete this document will have the conceptual foundation to engage with any document in the DBA-MA series without needing to reconstruct that foundation from the technical documents themselves.

2. How Nuclear Facilities Stay Safe

Nuclear power plants do not remain safe by accident, nor do they remain safe solely because of the physics of fission. They stay safe because of a layered system of engineered design, defined threat envelopes, and disciplined human operation — three mutually reinforcing elements that together define what the nuclear safety community calls defense-in-depth. Understanding how these three elements work, and how they depend on each other, is essential background for understanding why military action represents a categorically different kind of challenge.

2.1 The Design Basis Accident: Defining the Envelope

Every nuclear facility is designed around a specific set of postulated accidents — events that the plant must be capable of surviving without unacceptable radiological consequences. This set is called the design basis. The accidents within it are called Design Basis Accidents, or DBAs.

The DBA concept is not a pessimistic exercise in imagining disasters. It is a disciplined engineering methodology. Regulators and designers agree in advance on a defined envelope of credible events — the worst credible loss of coolant, the worst credible pipe break, the worst credible earthquake for that site — and the plant is then engineered to withstand each of those events with its safety systems intact. The DBA envelope drives safety system design. If a postulated accident falls within the envelope, the plant must be able to handle it. If an event falls outside the envelope, it is — by definition — beyond the design basis, and different analytical and regulatory treatments apply.

This approach has a powerful practical virtue: it gives designers, regulators, and operators a shared, precise language. When an engineer says “this event is within design basis,” everyone understands what that means — the safety systems were designed and tested for it, the operators have procedures for it, and the licensing basis accounts for it. When someone says “this is a beyond-design-basis event,” that too has a defined meaning: the plant may still be able to cope, but it will be doing so outside the envelope for which it was specifically designed, using capabilities that may not have been fully analyzed. The DBA framework is simultaneously a design driver, a regulatory commitment, and a common vocabulary.

2.2 The Design Basis Threat: The Security Analog

The security community developed a parallel concept for the physical protection of nuclear facilities: the Design Basis Threat, or DBT. The DBT defines the adversary that the physical protection system of a nuclear facility must defend against. Like the DBA, it is deliberately bounded. The DBT specifies adversary characteristics — number of attackers, level of sophistication, insider threat potential, and available weapons and tools — and the facility’s physical protection systems are then designed and tested against that defined threat. Guards, barriers, access controls, intrusion detection systems, and response protocols are all calibrated to the DBT.

National regulators set the DBT in consultation with intelligence and law enforcement agencies. In the United States, the NRC establishes the DBT for commercial nuclear facilities. The DBT is treated as sensitive security information — its specific parameters are not publicly disclosed — but its existence and structure are well understood within the industry.

The key structural point is this: the DBT, like the DBA, defines an envelope. Physical protection systems are required to defend against the defined threat. They are not required to defeat every conceivable threat. A nation-state military force conducting sustained operations against a nuclear facility is not within any DBT established for any commercial nuclear plant anywhere in

the world. It was never intended to be. The DBT was designed around the credible threats of sabotage, theft, and terrorism — not armed conflict between states.

2.3 Administrative Operating Procedures: The Human Layer

Design and threat envelopes are necessary but not sufficient. Between the engineered systems and the actual safe operation of the plant sits the human layer: the licensed operators, the technical specifications, and the administrative operating procedures that govern day-to-day and emergency operation.

Every licensed nuclear facility operates under a set of technical specifications — regulatory requirements that define the minimum conditions under which the plant may operate, the surveillance requirements that must be met to confirm safety systems are functional, and the action levels that require operators to shut down or take corrective action when conditions degrade. Technical specifications are enforceable regulatory commitments, not guidelines.

Underlying the technical specifications is a dense web of administrative procedures: normal, abnormal, and emergency operating procedures. Normal procedures govern routine operations — startup, shutdown, power maneuvering, and equipment surveillance. Abnormal procedures address off-normal conditions that do not yet constitute emergencies but require operator attention and response. Emergency operating procedures address accident conditions — they are written specifically to guide operators through DBA scenarios and their immediate aftermath.

Emergency operating procedures are not improvised. They are developed through systematic analysis of design-basis accidents, validated against plant simulators, and practiced by operator crews during regular training. When a DBA occurs, operators do not figure out what to do. They execute procedures that were developed, reviewed, and approved for exactly that event. This procedural infrastructure depends on several conditions being intact: operators must be present, trained, and able to communicate; safety systems must be functional or their degraded status must be known; external support — from the utility’s emergency response organization, from the regulator, from offsite emergency services — must be accessible; and the plant must be operating under conditions where the applicable procedures are the correct ones for the actual plant state.

2.4 How the Three Elements Work Together

In normal operations, these three elements function as a coherent system. The DBA defines the design envelope; the engineered safety systems implement the design response to that envelope; the operating procedures give licensed operators the tools to manage the plant through the envelope; and the DBT defines the security perimeter within which all of this takes place. Each element supports the others. Safety systems are only valuable if operators can activate and manage them — a requirement that entails procedures, training, and site access. Procedures are

only executable if operators are present and conditions are within the analyzed range. The DBT perimeter is only meaningful if the plant's operational integrity is maintained behind it.

The traditional DBA framework and the regulatory infrastructure that implements it rest on an implicit assumption: that the initiating event — whatever it is — is bounded in time, does not adapt in response to the plant's condition, and does not actively prevent the mitigation response. An earthquake does not modify its behavior because operators are trying to restore cooling. A pipe break does not deny access to the site. Equipment failures do not coordinate with each other to defeat multiple safety systems simultaneously. Military action violates every one of those assumptions. That is the gap this work is designed to fill.

3. The Gap: When the Assumptions Break Down

A state actor conducting military operations against a nuclear facility brings four characteristics that the traditional DBA and DBT frameworks were never designed to address.

The first is intentionality. The adversary chooses what to attack, when, and in what sequence. Unlike a natural hazard or an equipment failure, the adversary can observe the facility's response and adapt accordingly. If operators attempt to restore cooling, the adversary can prevent them from doing so. If an emergency generator is brought online, it can be targeted. The adversary is not indifferent to the outcome.

The second is persistence. The threat does not resolve on its own timeline. Zaporizhzhia has been under military occupation since March 2022. That is not an event with a defined duration; it is a sustained condition. The DBA framework was built around events — a pipe breaks, a pump fails, a tornado strikes. Military action can become an environment, one in which the facility must sustain safe operation indefinitely under conditions for which no operating procedure was written.

The third is precision. Modern military forces can target specific systems. A precision strike against a reactor protection system, an emergency feedwater pump room, or a spent fuel pool cooling system is not collateral damage — it is an attack designed to compromise a specific safety function. The traditional DBA framework postulates equipment failures; it does not postulate an adversary who has studied the facility's safety analysis report and is targeting the systems whose loss creates the most difficult recovery path.

The fourth is denial. Military action can prevent the mitigation response that the entire procedural infrastructure assumes will be available. Emergency operating procedures assume that operators can reach the control room, that repair crews can access damaged equipment, that external emergency response can reach the site, that replacement parts can be delivered, and that communications with the utility's emergency response organization and the regulator remain intact. Area denial tactics — sustained fires, blocked roads, armed perimeter control, air defense

suppression of helicopter access — can strip away every one of those assumptions without touching the reactor itself.

The consequence is not merely that the existing DBT is inadequate — though it is. The deeper problem is that the traditional DBA framework and the procedural infrastructure built on it become unreliable guides to plant behavior under conditions that violate their foundational assumptions. Operators executing emergency operating procedures written for equipment failures may find that the procedures address the wrong problem, or that the conditions required for successful procedure execution no longer exist.

The Design Basis Threat, as established for commercial nuclear facilities, focused on sabotage, theft, and terrorism. It was never intended to envelope a nation-state military force conducting sustained operations. This is not a regulatory failure; it reflects the considered judgment that commercial nuclear facilities are not military installations, and that their design basis should reflect the threats they credibly face in peacetime. That judgment was reasonable when it was made. Zaporizhzhia demonstrated that it is no longer sufficient as a complete answer.

4. The DBA-MA Series: Closing the Gap

The DBA-MA document series addresses this gap by doing for military action what the traditional DBA framework did for equipment failure and natural hazards: defining the threat envelope, establishing the analytical methodology for consequence analysis within that envelope, and providing the regulatory and operational tools needed to assess, prepare for, and mitigate the consequences.

The series is organized hierarchically, mirroring the parent-child structure of the nuclear regulatory framework itself.

4.1 The IAEA Parent Document

The L1 IAEA Framework establishes the methodology at the international level in terms that are regulator-neutral and applicable across any national licensing regime. It defines the threat categories that DBA-MA analysis must address: precision strike, area denial, sustained occupation, collateral damage from nearby operations, and cyber-kinetic combined attack. It establishes the consequence taxonomy and the core analytical methodology. It is positioned as an extension of existing IAEA safety standards — referencing the Safety Fundamentals, General Safety Requirements, and Specific Safety Requirements series — rather than a replacement for them. The DBA-MA methodology is a rigorous evolution of accepted practice applied to a new class of initiating conditions, not an ad hoc response to a single event.

4.2 National Child Implementations

National child documents inherit the methodology from the IAEA parent and adapt it to a specific country's regulatory framework, threat environment, facility inventory, and defense posture. The Finland and the United States national implementations are the first two children in the series. Finland is the template case: its threat environment is concrete and well-defined, its proximity to Russia makes the DBA-MA threat non-hypothetical, and its NATO membership creates the alliance coordination context that most European implementations will share. The United States' implementation addresses the world's largest commercial nuclear fleet under the NRC regulatory framework. It maps the DBA-MA methodology onto the existing licensing basis structure, with explicit attention to the inter-agency coordination framework — NRC, NERC, DOE, and DOD — required for national-level decisions.

4.3 Functional Documents

Supporting the L1 national documents is a structured set of functional documents. The L2 Accident Description defines the initiating conditions and consequence chains in the technical detail required for safety analysis. The L3 Trigger Framework establishes the graduated decision structure — analogous to the technical specifications that govern plant operations — that defines when and how a facility transitions from normal operations through elevated alert states to mandatory shutdown. The L4 Monitoring Guidance defines the intelligence and situational awareness requirements that support trigger framework decisions: what information must be available, through what channels, and with what timeliness to make those decisions coherently.

4.4 Addenda and Use Cases

Addenda address specific threat vectors and implementation challenges in depth that the parent documents cannot provide. The UxS framework now comprises two instruments: DBA-UxS (v1.1), the cross-sector foundational document establishing the uncrewed systems taxonomy, common threat characteristics, hardening design principles, and mandatory review cycle; and DBA-MA-UxS-NA (v1.1), the nuclear implementation annex that applies the cross-sector framework to nuclear facility safety-significant systems and defines the Autonomous Safe Shutdown Condition as the nuclear protected operational state. Together, they address the full family of uncrewed systems threats — uncrewed aerial systems (UAS), uncrewed surface vessels (USV), and uncrewed undersea vehicles (UUV) — including passive kinetic defense principles across all three domains and the intelligence-trigger framework governing the mandatory review cycle. The Physical Proximity Attack Addendum addresses ground-force approaches to the facility perimeter. The Transition Architecture Addendum is described in Section 5 below.

Use case documents apply the DBA-MA methodology to particular attack vectors, facility types, or consequence chains in scenario-specific detail. The Spent Fuel Pool Use Case traces the full consequence chain from loss of cooling through thermal progression to potential radiological release under conditions resulting from military action, demonstrating the methodology in practice and providing scenario-specific analysis that validates the analytical framework.

4.5 The Autonomous Safe Shutdown Condition

The conceptual anchor for the entire series is the Autonomous Safe Shutdown Condition (ASSC). This is the target state that DBA-MA engineering and procedures are designed to achieve and sustain: a facility configuration in which safe shutdown can be maintained without dependence on off-site power, external logistics support, or external emergency response. The ASSC concept applies what the DBA-MA series calls the local sovereignty principle: protective capability must be passive, or at a minimum autonomous, because the conditions that create the need for it are precisely the conditions under which external support cannot be assumed. The adversary who imposes military action on a nuclear facility is, by definition, also an adversary of the external support infrastructure that the facility's emergency response depends on.

4.6 The AI Governance Design Basis

A nuclear facility operating within the DBA-MA framework may simultaneously carry two distinct design bases that interact at the DBA level. The first is the physical DBA-MA design basis established by this series: the set of postulated conditions — equipment failures, natural hazards, and deliberate military action — that the facility must be designed and demonstrated to withstand. The second is an AI Governance Design Basis, established under the AI Governance series developed in parallel with this work, that governs the deployment of machine learning and generative AI systems in facility management, operational decision support, and safety-significant functions. As nuclear facilities increasingly incorporate ML-based state estimation, protection relay optimization, and operational analytics, and as Gen AI tools enter the operational support environment, the AI governance design basis becomes a required element of the facility's overall safety framework.

A denial condition that severs communications to a human Command Broker — the oversight authority required for actions above the Bright Line (FD-BL §4.5) — removes the governance layer at precisely the moment when the non-deterministic tools that generate those actions are most likely to be called upon under stress.

The governing principle is the Bright Line as defined in the Foundational Definition — The Bright Line (FD-BL v0.2). The Bright Line is a threshold on actions, not a property of a component: an action is above the line when its incorrect performance could produce a consequence at or above the facility's severity threshold, and requires substantive human authorization by a qualified Command Broker enforced architecturally (FD-BL §4.2, §4.5). Determinism enters only as the scope carve-out (FD-BL §4.4): components that are deterministic within a stated input bound and formally testable as deployed fall outside the Command Broker obligation, while generative or otherwise non-deterministic components carry it — but the carve-out is not the line, and a deterministic component can still produce an above-line action.

5. The Transition Architecture: Sequencing the Work

The DBA-MA methodology defines what full compliance looks like. The Transition Architecture addendum addresses the harder practical question: given that no existing facility has achieved full compliance, how does the industry get there without leaving the transition period itself analytically unmanaged?

This matters because the transition period is not a neutral interval. Every nuclear facility now within the scope of the DBA-MA framework is currently operating below the full DBA-MA design basis. The threat the framework was designed to address is not a future possibility; it is a present operational condition at Zaporizhzhia and an assessed risk at facilities across NATO's eastern flank. Managing the gap between the current posture and full compliance requires the same analytical discipline as defining the compliance standard itself. A framework that defines a destination without managing the journey leaves the most dangerous phase — partial implementation, when some measures are in place while others are not — without analytical oversight.

5.1 The Partial Implementation Hazard

A critical property of the DBA-MA framework is its systemic character: the engineering measures, the trigger framework, the monitoring infrastructure, and the governance authorities are mutually dependent. A facility that has implemented ASSC engineering measures but has not established the trigger framework has hardware that it cannot deploy under the decision criteria that should govern its deployment. A facility that has established monitoring feeds but has not integrated them into an assessed trigger framework has information it cannot act on coherently. Partial implementation does not yield proportional partial protection; in some configurations, it provides less protection than no implementation, because the presence of some measures may create a false sense of confidence that the actual posture does not warrant.

The transition architecture is therefore designed not merely to sequence gap closure, but to ensure that at each interim tier the implemented measures constitute a coherent, self-consistent protective posture. A facility operating at a defined tier should be able to describe exactly what that tier provides and, equally importantly, what it does not provide — so that the limitations of the interim posture are explicit, understood, and managed.

5.2 Three Interim Tiers

The Transition Architecture defines three interim tiers — TA-1, TA-2, and TA-3 — representing achievable, coherent protective postures at defined intervals on the pathway to full compliance, designated TA-F. Tiers are cumulative: each tier incorporates all requirements of the tiers below it.

TA-1 is the assessment and baseline tier. A facility at TA-1 has completed the foundational intellectual work: a rigorous facility-specific vulnerability assessment against each DBA-MA threat category; a gap analysis that immediately distinguishes safety-significant gaps from

significant-but-deferrable ones; a preliminary ASSC transition procedure; and an inventory of which monitoring feeds required by the L4 document are and are not accessible. TA-1 does not require any engineering changes. It requires that a facility know exactly where it stands and has committed that knowledge to a document that a regulator can review and accept. The logic is fundamental: you cannot prioritize what you have not measured. The vulnerability assessment and gap analysis at TA-1 produce the analytical basis for every subsequent decision about where to invest engineering resources and how to sequence implementation.

TA-2 closes the most immediately consequential gaps: those associated with loss of external power and access denial. A facility at TA-2 has on-site diesel fuel and consumables sufficient to sustain safety-critical systems for at least 72 hours — and for 7 days at facilities near active or potential military operations — without dependence on the commercial supply chain. It has a validated ASSC transition procedure, tested through a tabletop exercise involving operators, regulators, and national defense contacts. It has functioning intelligence and monitoring feeds. And it has established legal authority for the Tier 2 and Tier 3 shutdown decisions defined in the trigger framework. This last requirement cannot be deferred: a facility that can transition to the ASSC but lacks the decision authority to do so coherently has a capability it cannot deploy. The focus on loss-of-power and access-denial consequences at TA-2 — rather than hardening against direct precision strike — reflects the observation that these consequence conditions are common to all DBA-MA threat categories. Whatever the initiating military action, the consequence pathway that creates the greatest safety challenge is the same: loss of external power, loss of external logistics, loss of site access for repair and resupply. TA-2 closes those common consequence pathways first.

TA-3 brings the facility to full operational capability at the procedural level before the final engineering hardening is complete. At TA-3, the facility has demonstrated, through a full-scale exercise with national defense and emergency response organizations, that it can execute the complete Tier 1 through Tier 3 trigger sequence under simulated military conditions. It has extended autonomous operation capability to 30 days. It has completed a supply chain provenance review of safety-critical components, identifying dependencies on adversary-state supply chains that would be at risk in the conflict environment addressed by the DBA-MA framework. And it has completed a UxS threat assessment covering uncrewed aerial, surface, and undersea systems as required by DBA-UxS Section 9, and a passive kinetic defense design study across all applicable domains — the analytical work required to scope and cost the physical barrier measures that TA-F will require. The residual gap between TA-3 and full compliance is the physical hardening itself. TA-3 ensures that a facility arrives at that final engineering work with the assessment, procedures, monitoring, authority, logistics, and exercised operational capability already in place.

5.3 Schedule and Acceleration Triggers

Maximum intervals between tiers — 18 months to TA-1, 36 months from TA-1 to TA-2, 48 months from TA-2 to TA-3, and 60 months from TA-3 to TA-F — yield a total transition ceiling of approximately 13.5 years from framework entry into force to full compliance. That is a long horizon, and it is intended to be honest about the scale of the work rather than optimistic about what can be achieved quickly. The interim tiers ensure that meaningful protection accrues throughout that period rather than at the end.

The schedule acceleration triggers built into the Transition Architecture recognize that the threat environment may not cooperate with a 13.5-year timeline. A credible specific threat to a facility, military action against a nuclear facility anywhere in the world, or a material deterioration in the national threat environment — including significant deterioration of bilateral relations with a nuclear-armed state, or deployment of new adversary precision strike capabilities within range of domestic nuclear facilities — each requires immediate reassessment of whether the accepted milestone schedule remains adequate. These acceleration triggers parallel the explicit intelligence triggers in the DBA-UxS mandatory review cycle framework (Section 9), applying the same principle: that a defined methodology cannot be static in a dynamic threat environment.

6. Document Architecture at a Glance

The following describes the current document inventory of the DBA-MA series. All documents are in active development but are available for distribution to nuclear safety regulators, international bodies, and national defense planners.

6.1 Framework Documents

The L1 IAEA Framework (v1.7) is the parent document establishing the DBA-MA methodology at the international level. The L1 United States National Implementation (v2.6) and the L1 Finland National Implementation (v1.3) are the first two national implementations. The L2 Accident Description (v2.3), L3 Trigger Framework (v2.1), and L4 Monitoring Guidance (v2.3) are IAEA-level instruments applicable across all national implementations; jurisdiction-specific regulatory mappings and program details are addressed in standalone national child documents for each implementation. The L2 defines initiating conditions, consequence chains, and the five governing engineering principles. The L3 establishes the graduated decision structure governing facility response. The L4 defines the intelligence and situational awareness requirements supporting trigger framework decisions.

6.2 Addenda

The UxS framework comprises two instruments released concurrently: DBA-UxS (v1.1), the cross-sector foundational document, and DBA-MA-UxS-NA (v1.1), the nuclear implementation annex. Together, they address uncrewed systems threats across the aerial (UAS), surface maritime (USV), and undersea (UUV) domains, establish the uncrewed systems taxonomy,

define hardening design principles for each domain, and define the mandatory review cycle framework. DBA-MA-UxS (v1.0) is retired as a standalone instrument; references to it in earlier DBA-MA series documents shall be understood as referring to the DBA-UxS and DBA-MA-UxS-NA pair upon adoption. The DBA-MA-PPA Addendum (v1.1) addresses physical proximity attack scenarios. The DBA-MA Transition Architecture (v1.0) establishes the interim implementation tier framework described in Section 5 of this document.

6.3 Use Cases and Companion Documents

The DBA-MA-UC-1 Use Case (v1.0) traces the spent fuel pool thermal progression consequence chain under military action conditions. The Three Threads of Cyber Security (v1.1) addresses the cyber dimension of the DBA-MA threat environment. The Ukraine Grid Defense companion document addresses bulk electric system resilience under kinetic and cyber attack. The Electric Sector Design Basis Series provides the non-nuclear electric infrastructure consequence analysis: DBA-ES: Electric Sector Design Basis Framework (v0.6) is the sector-level parent; DBA-ES-HY-FC1 (v0.3) addresses large hydroelectric storage facilities; and DBA-ES-TX-FC2 (v0.1) addresses large EHV transmission substations. The Non-Nuclear Electric Power Infrastructure companion document (v1.0) is superseded and retired; the DBA-ES series instruments are the authoritative successors.

6.4 Positioning Within Existing Regulatory Frameworks

Every document in the DBA-MA series is positioned as an extension of, not a replacement for, the existing nuclear safety and security regulatory framework. The IAEA parent explicitly references the Safety Fundamentals, General Safety Requirements, and Specific Safety Requirements series. National implementations map DBA-MA requirements onto the host country's existing licensing framework. The methodology is designed to be adopted by regulators as an extension to existing safety assessment guidance — new territory, built on the foundation of decades of established practice.

Revision History

Version	Date	Author	Description
1.0	April 2026	T. Roxey	Initial release. Standalone orientation document introducing the DBA-MA framework, document architecture, core technical concepts, and transition pathway to full compliance.
1.1	April 2026	T. Roxey	Editorial revision. Grammarly review applied throughout.
1.2	April 2026	T. Roxey	Section 6.1 updated to reflect elevation of L2, L3, and L4 from US-scoped to IAEA-level instruments (v2.0 each) and updated version numbers for all L1 documents (IAEA v1.3, US v2.4, Finland v1.3). Cover version corrected.

1.3	April 2026	T. Roxey	Editorial revision. Grammarly review applied throughout.
1.4	April 2026	T. Roxey	Preface updated to incorporate HYDRA team details: the late August 2021 convening of Jeff R., Rob L., and Mark F., predating the February 6th 2022 letter to Ukrainian authorities.
1.5	April 2026	T. Roxey	Preface revised: HYDRA paragraph updated to add State Department context and correct name conventions; February 6th paragraph updated to reflect Embassy delivery of the letter.
1.6	April 2026	T. Roxey	Section 6.1 updated to reflect version increments across the series (IAEA L1 v1.4, US L1 v2.5, L2/L3/L4 v2.1) and extraction of national implementation annexes to standalone national child documents.
1.7	April 2026	T. Roxey	Sections 4.4, 5.3, and 6.2 updated to reflect restructuring of the UxS addendum into two instruments: DBA-UxS (v1.0), the cross-sector foundational document, and DBA-MA-UxS-NA (v1.0), the nuclear implementation annex. DBA-MA-UxS (v1.0) retired as a standalone instrument. Terminology corrected in all UxS-related passages: “uncrewed” now used consistently throughout.
1.8	May 2026	T. Roxey	Section 6.1 version references updated: IAEA L1 v1.4 → v1.6, US L1 v2.5 → v2.6, L2 v2.1 → v2.2, L4 v2.1 → v2.3. Section 6.3 electric sector entry replaced: Non-Nuclear Electric Power Infrastructure companion document superseded and retired; DBA-ES Electric Sector Design Basis Framework (v0.4) and DBA-HY-FC1 (v0.3) named as current series instruments. Addresses Gap 2.3 of DBA Series Gaps and Recommendations v1.2.
1.9	June 2026	T. Roxey	Section 4.6 added: The AI Governance Design Basis. Establishes the dual-DB architecture for nuclear facilities carrying both a physical DBA-MA design basis and an AI Governance Design Basis; explains their interaction at the DBA level under military action conditions; introduces the Brightline and Command Broker requirements as architectural elements that must be preserved through ASSC engineering measures. Section 6.1 version references updated: IAEA L1 v1.6 → v1.7, L2 v2.2 → v2.3. Section 6.2 version references updated: DBA-UxS v1.0 → v1.1, DBA-MA-UxS-NA v1.0 → v1.1, DBA-MA-PPA v1.0 → v1.1. Section 6.3 electric sector entry updated: DBA-ES advanced to v0.6; DBA-HY-FC1 renamed to DBA-ES-HY-FC1 (v0.3); DBA-ES-TX-FC2 (v0.1) added as new series instrument.